



Full length article

Ovicidal, larvicidal and oviposition repelling action of a nanoemulsion of citronella essential oil (*Cymbopogon winterianus*) on *Cochliomyia hominivorax* (Diptera: Calliphoridae)

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ABSTRACT

Cochliomyia hominivorax is the dipteran that causes the worst and most serious types of myiasis in animals and humans in the Americas. They are parasitic affections caused by the massive infestation of larvae of this species on the tissues of living mammals. The routine treatment uses chemosynthetic products that have high toxicity and can cause several injuries to animal and human health, besides environmental contamination. The development of more friendly alternatives to the treatment of this disease is an important knowledge gap nowadays. Citronella (*Cymbopogon winterianus*) is the most widely cultivated medicinal plant in tropical and subtropical regions. Essential oils have a recognised repellent and larvicidal activity against a wide variety of insects. This work aimed to evaluate the repellent, ovicidal and larvicidal efficacy of a topical commercial formulation composed of 15% (w/v) nanoemulsified Citronella essential oil (Nano Citronela Plus®) in *in vitro* trials on specimens of *C. hominivorax* created under laboratory conditions. Nano Citronela Plus® is an oil-in-water nanoemulsion obtained by high-pressure homogenization method. The composition of tested nanoemulsion was *Cymbopogon winterianus* essential oil, vanillin, caprylic/capric triglyceride, polysorbate 80, sorbitan oleate, tocopheryl acetate, sodium benzoate and water. Nano Citronela Plus® demonstrated 100% lethality on eggs and first and second stage larvae of *C. hominivorax*. Mortality in the larvae of third stage was 69.33% and inhibited the oviposition of *C. hominivorax* females. Nano Citronela Plus® proved to be a possible herbal alternative for the control and treatment of myiasis caused by *C. hominivorax*. *In vivo* studies may test this hypothesis.

Introduction

The fly *Cochliomyia hominivorax* is a dipteran of the order Calliphoridae (Coquerel, 1858) that causes myiasis in animals and humans. Currently, the species is known as “New World Screw-Worm” due to its endemic occurrence in tropical, subtropical and temperate zones of the western hemisphere (Hall and Wall, 1995; OIE, 2008; Costa-Júnior et al., 2019). *C. hominivorax* is present in all countries in South America, except Chile. This fly is also present in Cuba, Dominican Republic, Haiti, Jamaica and Trinidad and Tobago. *C. hominivorax* was eradicated from

the United States of America, Mexico, Guatemala, Belize, Honduras, El Salvador, Nicaragua, Costa Rica and Panama. (Wyss, 2000; Costa-Júnior et al., 2019).

According to the most recent version (2018) of the International Statistical Classification of Diseases and Health-Related Problems (ICD 11), myiasis is a tissue disease caused by an infection with fly larvae of the order Diptera. The World Health Organization classifies human myiasis as a serious medical condition that requires urgent treatment after its diagnosis (WHO, 2018). Myiasis represents a major health risk for people with inadequate hygienic habits, psychiatric problems and

Abbreviations: B.O.D., Biochemical Oxygen Demand; cm, centimeter; °C, centigrade degree; *C. hominivorax*, *Cochliomyia hominivorax*; DNA, deoxyribonucleic acid; DLS, Dynamic Light Scattering; g, gram; h, hour; IGR, insect growth regulator; Kg, kilogram; L, liter; L1, first stage larvae; L2, second stage larvae; L3, third stage larvae; mL, milliliter; mm, millimeter; nm, nanometer; EO, essential oil; R\$, Brazilian real; v, volume; µg, microgram; US \$, American dollar.

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alcoholism and who are more predisposed to skin lesions, thus increasing their susceptibility to these infestations (Francesconi and Lupi, 2016). Myiasis caused by *C. hominivorax* affects both zootechnical and pet animals in Brazil (Dantas-Torres and Otranto, 2014). The main reports are in zootechnical animals (Alves-Branco et al., 2001), however, companion animals (Pets) are also commonly affected by this type of myiasis (Meinking et al., 2003). In South America, the main form of control of the *C. hominivorax* fly in livestock is the massive use of chemosynthetic insecticides (Borja, 2003). The sterile insect technique was developed over 60 years ago and the fly was eradicated in North and Central America. The costs of developing an eradication program in the endemic regions as the Caribbean islands and along the west coast of South America have prevented its implementation. (Concha et al., 2016).

The search for new botanical insecticides has been intensifying in the last decade (Pavela, 2015). One of the main motivators of this movement is the exhaustion of the use of chemosynthetic insecticides (Isman and Grieneisen, 2014). Plants can provide compounds capable of exerting several biological effects of interest to human use, including insecticides, larvicides and repellents (Isman, 2006; Mukandiwa et al., 2012; Seuling et al., 2019).

Citronella (*Cymbopogon winterianus*) is a medicinal plant grown in tropical and subtropical regions of America, Asia and Africa, popularly known for its insect repellent characteristic (De Mendonça et al., 2005). The essential oil (EO) is extracted from the leaves of Citronella and it has a repellent action against mosquitoes, in addition to being a low cost herbal medicine when compared to synthetic agents (Quintans-Júnior et al., 2008). *Cymbopogon winterianus* belongs to the Poaceae family with proven therapeutic and medicinal values. Various components of its oil are widely used as a source of perfumery, soap, cosmetics and flavorings industries worldwide (Simic et al., 2008). Citronella EO is popularly known for its repellent characteristic of culicid insects (De Mendonça et al., 2005). It also has many potential bioactive components of great pharmaceutical and medicinal significance that can act as anti-inflammatories and antioxidants (Leite et al., 2010). The Citronella EO are mainly a mixture of monoterpenes and sesquiterpenes containing mainly limonene, linalool, geraniol, elemol, geranyl acetate, α -bisafalol, citronellol and citronellal (Lorenzo et al., 2000). Larvicidal effect has also been demonstrated for certain species of mosquitoes (De Mendonça et al., 2005; Rodrigues et al., 2013). The Citronella EO also caused repellency to adults of *Callosobruchus maculatus* (Fabr) (Gusmão et al., 2013) and food deterrence and larval mortality to *Spodoptera frugiperda* (Labinas e Crocomo, 2002).

Despite the potential application as insect repellent, the use of Citronella EO is limited due to its high volatility and low water solubility. Among the alternatives to overcome these limitations, the nano-encapsulation of EO has been considered a promising strategy to reduce volatility, improving stability, water solubility and efficacy of essential oil-based products (Pedro et al., 2013; Bilal et al., 2014). Nano Citronela Plus® is a commercial topical formulation based on the nanoemulsion of Citronella EO. Besides Citronella EO, Nano Citronela Plus® also contains vanillin, a natural additive that has shown to increase the protection time of Citronella preparations against mosquitoes (Tawatsin et al., 2001; Kongkaew et al., 2011).

This work aimed to evaluate the larvicidal, ovicidal and repellent efficacy of the commercial product Nano Citronela Plus®, formulated from a nanoemulsion of Citronella EO (15% w/v) on first, second and third stage larvae, eggs and adult females of *C. hominivorax* in *in vitro* trials.

Material and methods

Nano Citronela Plus®

Nano Citronela Plus®, a nanoemulsion containing 15% (w/v) of Citronella EO (*Cymbopogon winterianus*), was donated by NanoScoping

Soluções em Nanotecnologia (Brazil). Nano Citronela Plus® is an oil-in-water nanoemulsion obtained by high-pressure homogenization method. The composition of the tested nanoemulsion was *Cymbopogon winterianus* EO, vanillin, caprylic triglyceride, polysorbate 80, sorbitan oleate, tocopheryl acetate, sodium benzoate and water. The 15% (w/v) concentration was based on the highest concentration that resulted in a nano formulation with adequate size, homogeneous and stable characteristics.

Particle size

The size distribution, mean particle size, and polydispersity index (Pdl) of Citronella EO nanoemulsion were determined by dynamic light scattering (DLS), using a Zetasizer Nano ZS90 (Malvern Instruments, Worcestershire, UK). The measurements were made at 25 °C after appropriate dilution of the nanoemulsion in purified water. Size analyses were performed at a fixed scattering angle of 90°.

Cochliomyia hominivorax colony in laboratory

C. hominivorax laboratorial colony was kept according to the methodology described by Seugling et al., (2019). After the tests, the adult flies were kept under the above-mentioned conditions until they reached the end of their life cycle and died naturally. Normatives Directive 86/609/EEC on the Protection of Animals Used for Experimental and Other Scientific Purposes were followed.

In vitro test for the evaluation of ovicidal efficacy on *Cochliomyia hominivorax*

The ovicidal efficacy test used in this study was based on the methodology exposed by Showler et al., (2019). Methodological adaptations were made in order to adapt the ovicidal test reported by the authors to the *C. hominivorax* peculiarities. This test consisted of submitting a group of 15 eggs of *C. hominivorax* to direct contact with the tested substances and of observing this effect on the hatching of first stage larvae, 24 h after this contact. Initially, a 4 cm diameter Whatman® 125 mm filter paper disc was placed at the bottom of each vial (9 cm high × 4 cm in diameter). A group of 15 *C. hominivorax* eggs was fixed on each of the filter paper discs with the help of blunt forceps and binocular stereomicroscope. On each disc, 1 mL of the tested substances was sprayed in liquid form. Subsequently these devices were submitted to a Biochemical Oxygen Demand (B.O.D.) Newlab® incubation chamber with 36.5 ± 0.5 °C and 60% relative humidity for 24 h. After this period, the eggs hatching was measured with the aid of a binocular stereomicroscope. Each device containing a group of 15 eggs after the application of one of the tested products was considered to be an experimental unit. The average mortality rate for each treatment was calculated by the sum mean of the individual mortality rates for each experimental unit weighted by the number of treatment replicates. The experimental groups were: Nano Citronela Plus®, Distilled water (negative control group), Triclorphon synthetic insecticide (Neguvon® Bayer) in a concentration of 0.194% (v/v) or 1.94 g.L⁻¹, which is the indicated concentration by the manufacturer for veterinary topical use, to combat ectoparasites (positive control) and tests were also carried out with the vehicle of the Citronela Plus®, in order to ensure that the effect would be due to the essential oil, and not to the other components of Nano Citronela Plus®. Eight replicates were carried out for the group: distilled water, vehicle Nano Citronela Plus®, Triclorphon (Neguvon®) and Nano Citronela Plus®.

In vitro test for the evaluation of the larvicidal efficacy with first and second stage *Cochliomyia hominivorax* larvae

This test basically consisted of submitting groups of 15 larvae of first and second stage of *C. hominivorax* to direct contact with the tested

substances in its aqueous form and subsequent observation of the effect of this treatment on the continuity of the larval development. The first and second stage larvae of *C. hominivorax* were removed from the growth medium and their stage of larval development was identified with the aid of a binocular stereoscope. The followed morphological criteria were those proposed by Spradbery (1991). The implementation of these trials involved performing the contact test and a plastic container to keep the larvae in the growth medium after contact with the tested products. Initially, a 4 cm diameter Whatman® 125 mm paper filter disc was placed in the glass vial. A group of 15 larvae of first and second stage of *C. hominivorax* was placed on this filter paper and sprayed 1 mL of the substance in question, directly on them. After the application of the experimental treatment, the larvae remained in this environment for 1 min and then, they were sent to a second container, previously prepared with a 50 g layer of the mixture of ground beef and artificial liquid diet. After receiving the treated larvae, this recipient was kept in B.O.D. Newlab® incubation chamber, with temperature of $36.5 \pm 0.5^\circ\text{C}$ and 60% relative humidity for 24 h. After this period, the larvae viability was measured with the aid of a binocular stereoscope. The larvae were classified as alive or dead, according to their mobility and size in mm. Dead larvae are reduced in size when compared to alive ones, due to the interruption of their feeding and growth. Each device containing a group of 15 larvae was considered as an experimental unit. The average mortality rate for each treatment was calculated by the sum mean of the individual mortality rates for each experimental unit weighted by the number of treatment replicates. The experimental groups were the same as in the ovicidal test. Five replicates of each treatment were performed with first and second stage larvae. Changes in the color of the larvae exposed to contact with Nano Citronela Plus® were observed. These changes were recorded.

In vitro test for the evaluation of larvicidal efficacy with third stage *Cochliomyia hominivorax* larvae

Larvicidal efficacy tests were developed following the methodology by Seugling et al., (2019). The experimental groups were the same as in the ovicidal test. Twenty replicates of each treatment were carried out. Morphological abnormalities were observed after the pupation period in several of the specimens that came into contact with Nano Citronela Plus®. These malformations were recorded.

Test for the evaluation of repellency or oviposition inhibition on females of *Cochliomyia hominivorax*

The repellency test basically consisted of exposing adult flies to indirect contact with the tested substances in aqueous form and of observing the effect of this treatment on their oviposition. The performed evaluation computed in the presence or absence of oviposition and the weight of the egg masses was obtained in each treatment. The objective of this test was to simulate ideal conditions for oviposition of *C. hominivorax* in the presence of the tested substances and thus, ascertain whether they have the power to inhibit *C. hominivorax* oviposition. Falcon® tubes with capacity for 50 mL containing cotton were used. At the opposite end, a portion of 2 g beef was fixed. After the construction of this equipment, 1 mL of the tested substance was placed on the cotton and then a group of 10 female flies, 7 days old and healthy, was confined in this. The devices were then closed with a piece of voile-type fabric held by an elastic band and allocated to B.O.D. New Lab® incubating chambers with temperature of $36.5 \pm 0.5^\circ\text{C}$ and 60% relative humidity. After 1 h, the test result was evaluated by counting the number of ovipositions occurred in each tube and by weighing the egg masses. Each egg mass was considered to be an oviposition. The experimental groups were the same as in the ovicidal test. Each device with a group of 10 flies was considered to be an experimental unit. 10 replicates were performed for each experimental group. The mortality of *C. hominivorax* females submitted to the test was also analyzed.

Experimental outline and analysis of applied statistics

The considered response variable was the rate of mortality for the tests for ovicidal and larvicidal efficacy tests. For the repellency test, it was decided to use the number of ovipositions and their weight as a response variable. In the analysis of the observed mortality on adults exposed to the repellency test, the mortality rate was used as the response variable. Tests of adhesion to Normal distribution were performed on the residues of the linear model. Tests for homogeneity of variances among experimental treatments were performed. For all the developed tests, the variation factors were the experimental treatments: distilled water, vehicle of Nano Citronela Plus®, Nano Citronela Plus® and Triclorphon®.

As there was no normality about the residues and no homogeneity of variances was found, it was decided to apply the Kruskal-Wallis non-parametric statistical test (Kruskal and Wallis, 1952) in order to show the effect of treatments on mortality rates in the tests for ovicidal and larvicidal efficacy and on the weight of the eggs and number of ovipositions in the repellency test. The applied level for error probability was 5% ($p < 0.05$). The test *pos hoc* by Dunn (Dunn, 1964) was applied to separate the means. Data were analyzed by using the computational package SSPS® version 25.

Results

Particle size determination

According to dynamic light scattering (DLS) studies, Nano Citronela Plus® displayed a monodisperse distribution of particles ($\text{Pdl} < 0.2$), with a mean particle size of 203 nm.

Test for the evaluation of the ovicidal efficacy

Both Nano Citronela Plus® and the synthetic insecticide Triclorphon were lethal in all replicates of ovicidal efficacy trials using *C. hominivorax* eggs. The mortality observed in eggs in the distilled water group was 3.33% and in the vehicle group 4.17%. The observed means for Nano Citronela Plus® and Triclorphon groups were identical ($p = 1$). The means of the distilled water and Nano Citronela Plus® vehicle groups didn't differ ($p > 0.05$). The average mortality rates of *C. hominivorax* eggs submitted to experimental treatments are presented in Fig. 1.

Test for the evaluation of larvicidal efficacy

Both Nano Citronela Plus® and the commercial insecticide Triclorphon were lethal in all replicates of larvicidal efficacy tests with first and second stage *C. hominivorax* larvae. The mortality rate of first stage larvae observed in the distilled water group was of 14.67% and in the vehicle group 10.67%. The mortality rate of second stage larvae observed in the distilled water group was of 5.33% and in the vehicle group 6.67%. In both trials (L1 and L2) the averages observed for the Nano Citronela Plus® and Triclorphon groups were identical ($p = 1$) and the averages for the distilled water and Nano Citronela Plus vehicle did not differ ($p < 0.05$). The average mortality rate of first and second stage *C. hominivorax* larvae subjected to the experimental treatments are presented in Fig. 2.

While reading the results of the larvicidal efficacy trials on first and second stage larvae of *C. hominivorax*, some change in coloration of a large part of the larvae subjected to contact with Nano Citronela Plus® was observed. This phenomenon was observed in 43% of the specimens of this treatment. In Fig. 3, some specimens are presented with and without this finding, in order to allow comparison among them.

The mortality rate of third stage larvae observed in the distilled water group was of 6.67% and in the vehicle group 10.18%. The averages in the distilled water and Nano Citronela Plus® vehicle did not

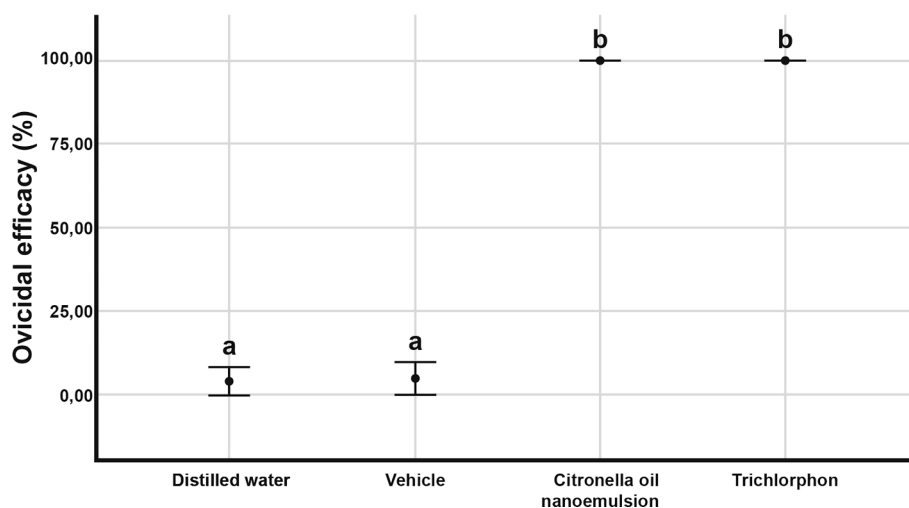


Fig. 1. Mortality rate (%) of *Cochliomyia hominivorax* eggs after contact with distilled water, Nano Citronela Plus® vehicle Nano Citronela Plus® and Triclorphon. The bars represent the confidence interval for the mean. Different letters signal statistically significant differences ($p < 0.05$).

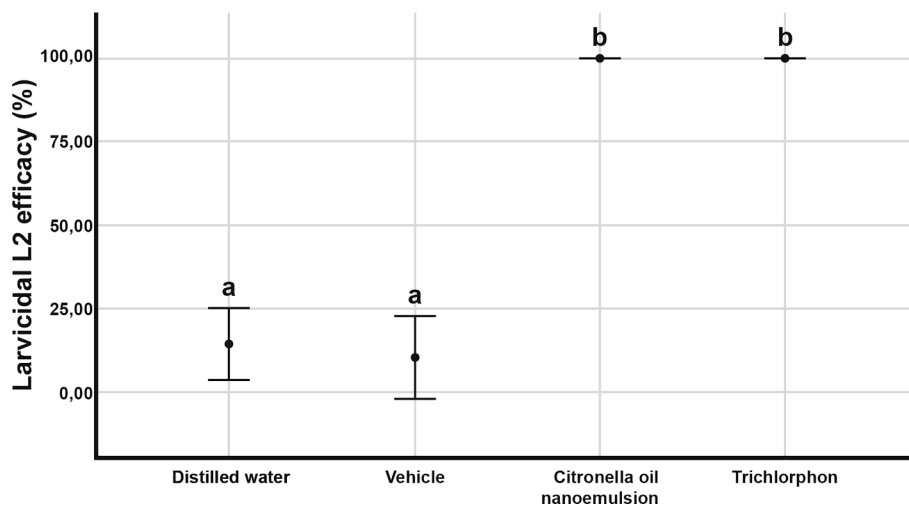
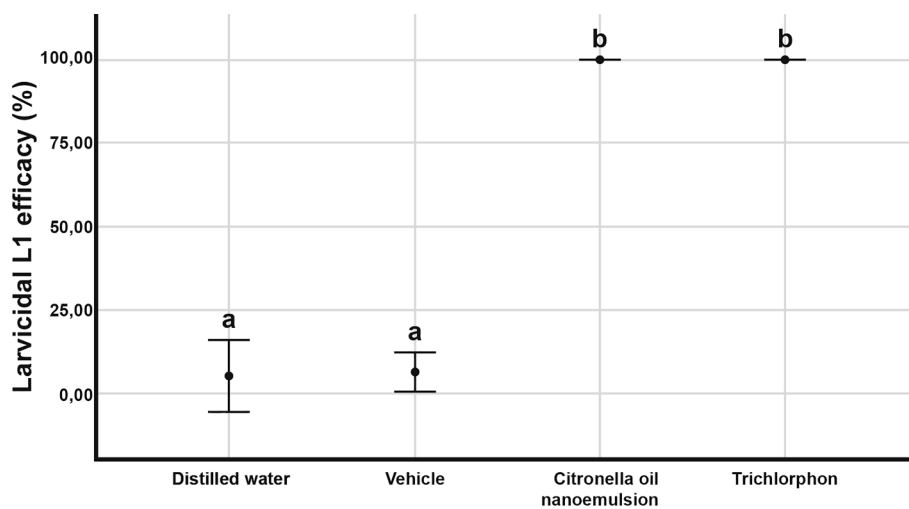


Fig. 2. Mortality rate (%) of first and second stage *Cochliomyia hominivorax* larvae after contact with distilled water, Nano Citronela Plus® vehicle, Nano Citronela Plus® and Triclorphon. The bars represent the confidence intervals of the mean. Different letters signal statistically significant differences ($p < 0.05$).



Fig. 3. Second stage *Cochliomyia hominivorax* larvae showing some change in characteristic color after contact with Nano Citronela Plus®. A: Larvae without the color change; B: Larvae with a blackened coloration.

differ ($p > 0.05$). The larvicidal potential from Nano Citronela Plus® over *C. hominivorax* was evidenced in this trial. The tested product stopped the development of 69.33% of third stage *C. hominivorax* larvae. Dunn's post hoc statistical test showed that this rate was significantly higher than that performed by the control groups ($p < 0.05$). As for the group treated with the synthetic insecticide Triclorphon, the observed mortality rate was 91.33%. The average mortality rates of third stage *C. hominivorax* larvae submitted to experimental treatments are shown in Fig. 4.

Morphological abnormalities were observed after the pupation period in some of the specimens that came into contact with Nano Citronela Plus®. These malformations are shown below in Fig. 5.

The upper part of Fig. 5 presents some malformations occurred in specimens that died during the pupal period. These failed to complete their development successfully. This phenomenon was observed in approximately one third of the larvae that died after contact with Nano Citronela Plus®. The lower part of Fig. 5. shows the malformations that occurred in the flies that managed to complete the development process. Specimen A shows severe malformation in the frontal region. This phenomenon occurred in approximately 10% of the larvae that died after contact with Nano Citronela Plus®. The B and C specimens have malformed wings to different degrees. This phenomenon occurred in approximately 50% of the specimens that died after contact with Nano Citronela Plus®. This was the most common type of malformation in this trial. Fig. 5 also shows some specimens without the occurrence of

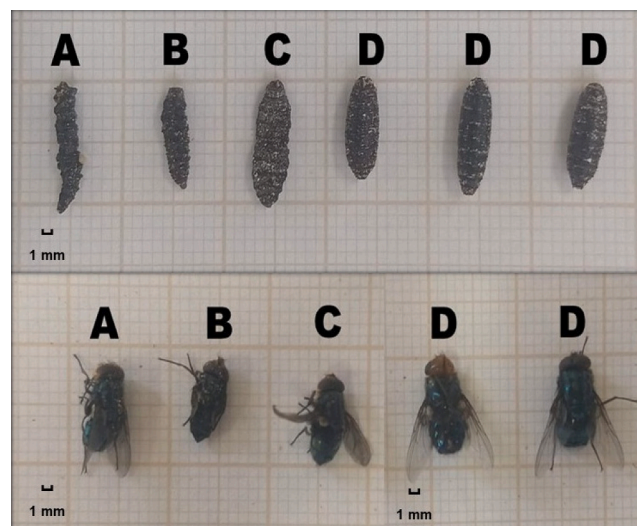


Fig. 5. Morphological abnormalities found in *Cochliomyia hominivorax* pupae and flies after contact of third stage larvae with a Nano Citronela Plus®. Top figure: A: Pupa with larviform aspect; B: Pupa with reduced size; C: Pupa with aberrant conformation; D: Pupae without malformations. Lower figure: A: Fly with atrophied head; B: Fly with reduced body size and absent wings; C: Fly with malformed wings; D: Flies without malformations.

malformations in order to provide comparison.

Test for the evaluation of the repellency

A total of 40 egg masses from flies subjected to the repellency test were counted. Out of these, 13 occurred in the distilled water group, 14 in the Nano Citronela Plus®, 7 in the Nano Citronela Plus® group and 6 in the synthetic insecticide group Triclorphon. As for the weight of these ovipositions, for the distilled water group a total of 0.2223 g was obtained, in the Nano Citronela Plus® vehicle group the egg weight was 0.2189 g, in the Nano Citronela Plus® group the weight of the egg masses was 0.1198 g and in the synthetic insecticide group Triclorphon 0.1212 g. Data are presented grouped in Table 1. There was a statistically significant difference ($p < 0.05$) for the effect of treatments on the oviposition count obtained in this trial. The egg mass weight variable did not show any significant difference among treatments. However, this may be due to the small number of replicates performed in the present

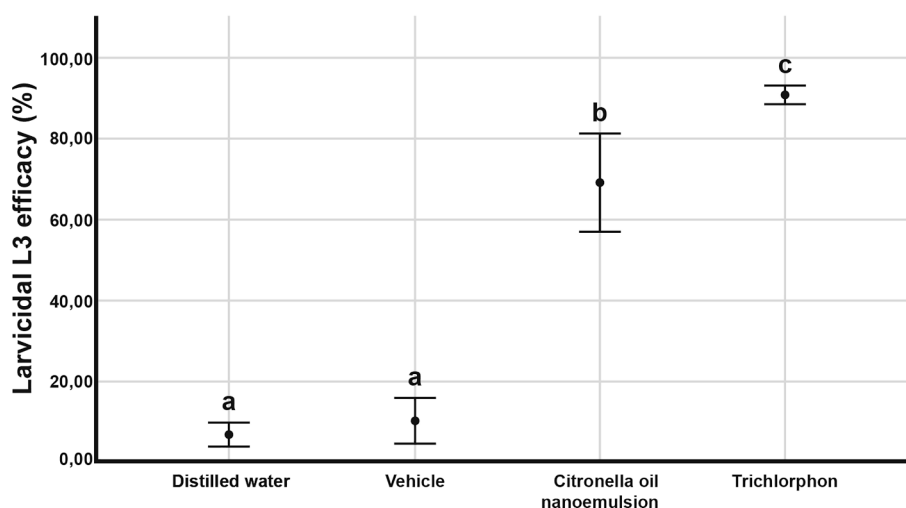


Fig. 4. Mortality rate (%) of *Cochliomyia hominivorax* third stage larvae after contact with distilled water, Nano Citronela Plus® vehicle, Nano Citronela Plus® and Triclorphon. The bars represent the confidence intervals of the mean. Different letters signal statistically significant differences ($p < 0.05$).

Table 1

Number of ovipositions, rate of oviposition inhibition (%) and *Cochliomyia hominivorax* eggs weight obtained after contact with Distilled water, Nano Citronela Plus® vehicle, Nano Citronela Plus® and Triclorphos. Different letters in the same column mean statistically significant differences ($p < 0.05$).

Treatment	Number of ovipositions	Rate of oviposition inhibition (%)	Eggs weight (g)
Distilled water	13 a	0,0 a	0,2223 a
Vehicle	14 a	0,0 a	0,2189 a
Nano Citronela Plus®	06b	42,85b	0,1198 a
Triclorphos	07b	50,0b	0,1212 a

study because the numerical difference found is biologically considerable.

The mortality of female *C. hominivorax* was also analyzed in this trial after oviposition. The negative controls of Nano Citronela Plus® vehicle and distilled water didn't present mortality. The mortality rates observed for Nano Citronela Plus® and the insecticide Triclorphos differed statistically ($p < 0.05$).

Discussion

Nano Citronela Plus® demonstrated considerable ovicidal, larvicidal and repellent powers over *C. hominivorax* in *in vitro* trials carried out in this work.

As for the larvicidal effect of Nano Citronela Plus®, this showed a very promising initial result in the tests performed. Lethality was 100% on eggs and young larval instars and approximately 70% efficacy on third stage larvae. As the present study deals with an *in vitro* trial, its preliminary character is evident. The technical guides published by the Working Group on the Effectiveness of Veterinary Medicines (European Commission III / 3682/92-EN) suggest that the ideal effectiveness for ectoparasiticides (natural or synthetic) in the treatment of diptera infestations be between 80% and 100%, and preferably, >90% (Committee for Medicinal Products for Veterinary Use, 1993). As the next action on research and development on Nano Citronela Plus®, investing in technologies able to increase its larvicidal efficacy is suggested. The topic of prospecting for botanical insecticides has currently received great attention from researchers (Isman, 2015). Citronella (*Cymbopogon winterianus*) is a popularly used plant for the control of ectoparasites in traditional veterinary medicine (Mello-Peixoto et al., 2013). This plant is well known for its repellent effects, especially against flies and horseflies (Tabanidae) (Nerio et al., 2010). However, there are still no reports of its use in combating the *C. hominivorax* fly.

The EO have been thoroughly researched related to flies control (Isman, 2020). Recently, Benelli et al., (2020) have highlighted the repellent powers of *Hazomalania voyronii* EO on *Aedes aegypti* (Diptera: Culicidae) and on *Culex quinquefasciatus* (Diptera: Culicidae). The *Aedes buchneroides* EO has also demonstrated repellent effect on *Aedes albopictus* (Diptera: Culicidae) (Ma et al., 2019). Benelli et al. (2019), Benelli et al. (2020) reported larvicidal effect of EO on the *Musca domestica* (Diptera: Muscidae). As for the dipterans of the Calliphoridae family, the larvicidal effect of up to 97% of the nanoemulsion of the *Baccharis dracunculifolia* OE on *C. hominivorax* third stage larvae in *in vitro* trials similar to the ones carried out in the present study were reported (Seugling et al., 2019). The effect of the non-emulsified *Baccharis dracunculifolia* EO has also been tested by Chaaban et al. (2018a), Chaaban et al. (2018b) with 71% of larvicidal action over *Cochliomyia macellaria* (Diptera: Calliphoridae). The *Piper gaudichaudianum* EO has also shown larvicidal activity over *Lucilia cuprina* (Diptera: Calliphoridae) (Chaaban et al., 2018b) and the *Curcuma longa* EO over *Cochliomyia macellaria* (Diptera: Calliphoridae) (Chaaban et al., 2019).

Several studies have been developed with the objective of elucidating the mechanisms of action of botanical insecticides (Enan, 2005; Rattan, 2010; Pavela and Benelli, 2016). One of the mechanisms of

action of botanical insecticides already known is the interference in the delicate cellular and humoral signaling processes that occur during the process of insect development (Pavela, 2016; Isman, 2020). This mechanism is well described in botanical insecticides based on azadirachtin, an important regulating factor for insect growth (IGR) (Isman, 2006). The occurrence of morphological anomalies and malformations in adult specimens (Fig. 5) may be linked to this mechanism of action, since Khater et al., (2011), Chaaban et al., (2018b), and Seugling et al. (2019) demonstrated similar effects in their studies that involved *in vitro* trials that promoted the contact of Diptera larvae with EO.

Nano Citronela Plus® demonstrated a repellent/oviposition inhibitory action on fertile females of *C. hominivorax*. Exposure to the product was able to significantly inhibit the oviposition of individuals submitted to this experimental treatment. The oviposition count obtained in this test was identical between the Nano Citronela Plus® group and the synthetic insecticide Triclorphos ($p > 0.05$). Triclorphos is an organophosphate traditionally used in commercial chemosynthetic products for the treatment of myiasis in livestock animals.

There may have been changes in the internal structures of adult insects that are imperceptible to macroscopic analysis. There may be a possibility that individuals which have managed to resist contact with Nano Citronela Plus® have suffered internal damage that is imperceptible to our tests, such as impaired fertility. This would be another form of control of the populations of this dipterous performed by the product and which should be evaluated in the next stages of this area of research. Further studies will rely on the evaluation of the reproduction of specimens surviving contact with Nano Citronela Plus® in order to answer this question.

The study of products derived from plants has developed a lot in the last decades. The commercialization of new botanical insecticides has increased considerably. In this context, the study of EO is still recent, but very promising. Botanical insecticides will be fundamental for the success of agriculture in developing countries and tropical regions, due to the great diversity of plants and the dangers of pesticides to the environment (Isman, 2020). Botanical insecticides will be likely substitutes for pesticides in the future.

Conclusion

Nano Citronela Plus® demonstrated considerable ovicidal, larvicidal and repellent effects over *C. hominivorax* and it represents an interesting promise in the prospect of new botanical insecticides for prevention and treatment of myiasis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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